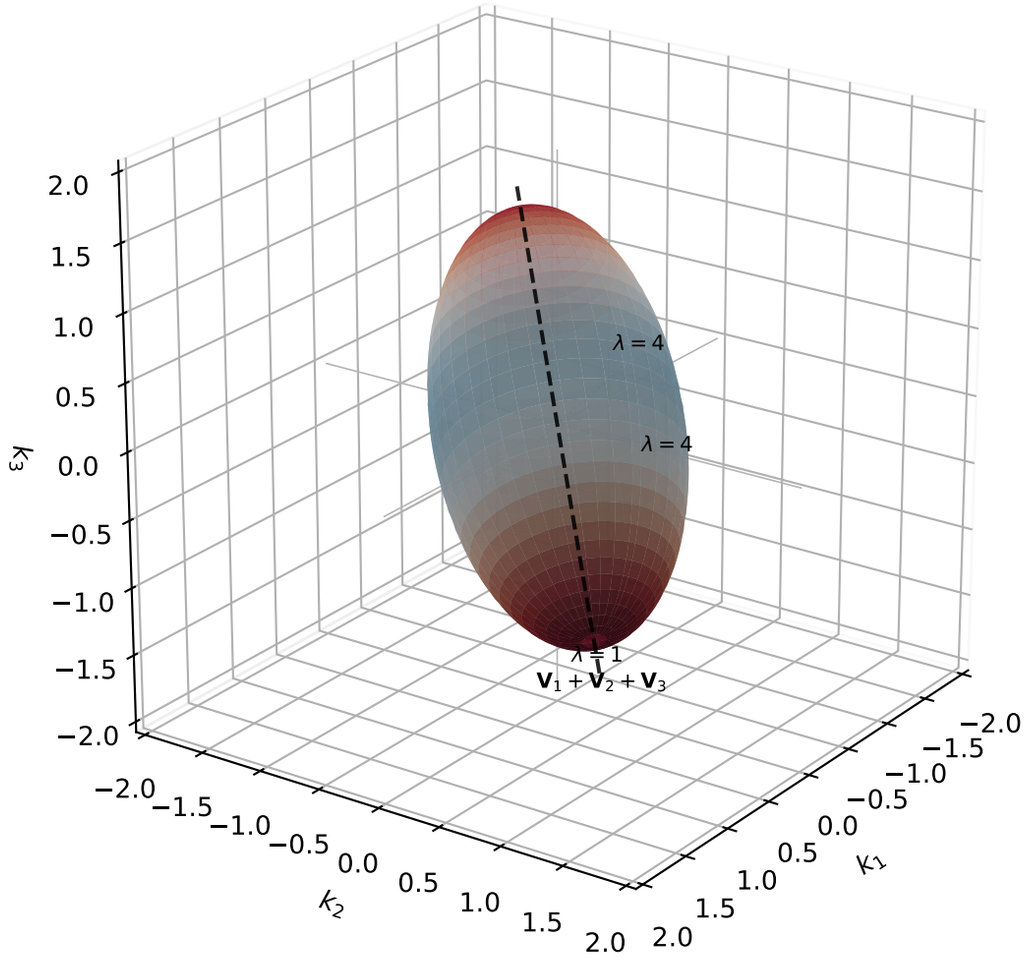
Dual anisotropy: kinematic ($M$, $k$-space) and gauge sector ($\mathbf{Q}$, $B$-space)
share the same optical axis $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 = (1, 1, -1)$

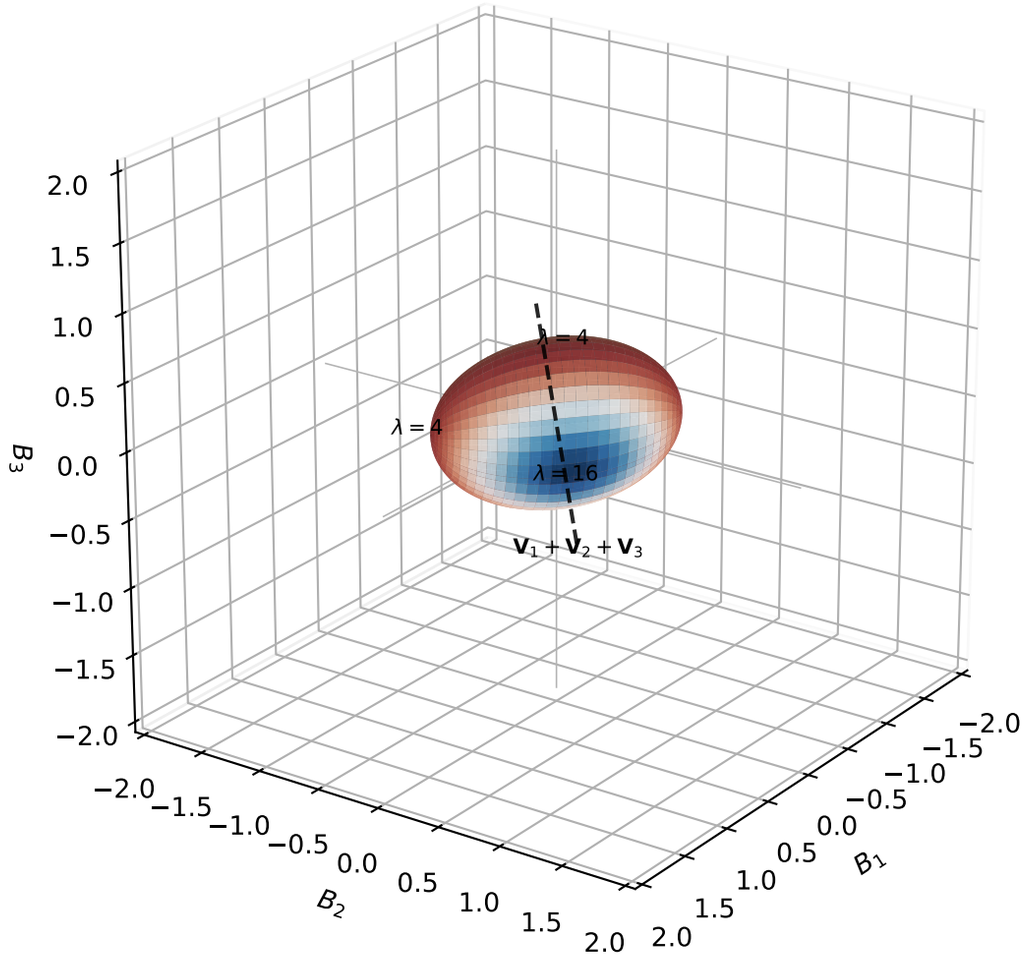
Kinematic sector: $M = \sum_{\text{RGB}} \mathbf{v}\mathbf{v}^T$

eigenvalues $\{4, 4, 1\} \Rightarrow$ prolate (**long** axis along $(1, 1, -1)$)

Gauge sector: $\mathbf{Q}$ from bipartite plaquette sum (App. B)

eigenvalues $\{4, 4, 16\} \Rightarrow$ oblate (**short** axis along $(1, 1, -1)$)




Both surfaces are level sets at $x^T A x = 3$. In the kinematic sector ($A = M$, eigenvalues $\{4, 4, 1\}$) the long ellipsoid axis lies along $(1, 1, -1)$: dispersion is *flat* along the optical axis and *steep* in the perpendicular plane. In the gauge sector ($A = \mathbf{Q}$, eigenvalues $\{4, 4, 16\}$) the short axis lies along $(1, 1, -1)$: the photon kinetic term is *steepest* along the axis and *flatter* perpendicular -- the dual phenomenology. Same geometry, dual shapes; one optical axis, two falsifiable signatures (P7 plus the gauge-sector birefringence in App.~B).